

KHALSA COLLEGE OF ENGINEERING AND TECHNOLOGY

Affiliated to IK Gujral Punjab Technical University, Punjab

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DEPARTMENT OF COMPUTER APPLICATION

PROJECT REPORT

ON

“ONE STOP SOLUTION”

In Partial fulfilment of the requirement for the degree of

BACHELOR OF COMPUTER APPLICATION

SUBMITTED BY:

Shailesh Kumar [2304050]

SUBMITTED TO:

Er. Sukhmeet Kaur

CERTIFICATE

This is to certify that the project entitled “One Stop Solution”, submitted by **Shailesh Kumar**, is a Bonafide work carried out under the guidance of **Mr. Vishal Singh, Ms. Shivani Mahajan** in partial fulfillment of the requirements for the award of the degree in Bachelor of Computer Application (BCA) from Khalsa college of engineering & technology.

The work presented in this project report is original and has not been submitted elsewhere for any other degree or diploma. The project has been completed as per the guidelines prescribed by the institution and demonstrates the students’ understanding of the subject.

We hereby approve this project as a successful completion of the academic requirements.

Signature of Student

Signature of Guides

Ms. Shivani Mahajan

Mr. Vishal Singh

Signature of Head of Department

Date:_____

Place:_____

DECLARATION

I, **Shailesh Kumar**, hereby declare that the project titled is a “**One Stop Solution**”, is carried out by me and my friend (**Aditya**) for the degree of Bachelor of Computer Applications (**BCA**) and submitted in Bachelor of Computer Applications, Khalsa College of Engineering and Technology, Amritsar, is a record of an original work done by me my friend under the guidance of **Mr. Vishal Singh** .

I further declare that this project has not been submitted to any other university or institution for the award of any degree/diploma. The work presented in this report is based on my own efforts.

I take full responsibility for the authenticity and correctness of the information presented in this project report.

Date: _____

Place: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my respected project guides, **Mr. Vishal Singh, Ms. Shivani Mahajan** their valuable guidance, continuous support and encouragement, constructive suggestions throughout the development of my project titled is “**One Stop Solution**” is a Centralized Event Website.

I am also highly thankful for our Head of the Department, **Er. Sukhmeet Kaur**, for her motivation and for providing the necessary facilities to carry out this project work.

I extend my heartfelt thanks to the management and faculty of Khalsa College of Engineering Technology, Amritsar, for giving me the opportunity to work on this major project, which enhanced my technical knowledge and practical skills in web development, database management, and system design.

Lastly, I would like to thank my parents and friends for their constant encouragement and moral support throughout this journey.

Date: _____

Aditya (2303957)

Place: Amritsar, Punjab

Department of **(BCA)**

Content

Sr. No	Description	Pg. No
	ABSTRACT	1
	Chapter 1: INTRODUCTION	2 – 4
	Chapter 2: LITERATURE REVIEW	5 - 6
	Chapter 3: PROBLEM STATEMENTS	7 - 8
	Chapter 4: SYSTEM ANALYSIS	9 – 11
	Chapter 5: SYSTEM DESIGN	12 – 21
	Chapter 6: SOFTWARE & HARDWARE	22
	REQUIREMENT	
	Chapter 7: IMPLEMENTATION	23 - 35
	Chapter 8: TESTING	36 - 37
	Chapter 9: RESULTS & DISCUSSION	38 - 42
	Chapter 10: FUTURE SCOPE	43 - 44
	Chapter 11: CONCLUSION	45
	Chapter 12: REFERENCES	46
	APPENDIX	47 - 50

ABSTRACT

The project titled “**One Stop Solution**” is a web-based centralized event, conference, workshop management system designed to connect multiple colleges in Amritsar on a **single digital platform**. The main objective of the system is to overcome the problem of scattered event information and provide an efficient way to manage and participate in **academic and non-academic events**.

This system improves **communication** between colleges, increases **student participation**, and provides a structured and scalable solution for event workshops conference management. The project demonstrates practical implementation of web development, database management, and system design concepts.

The system allows different types of users such as **students, faculty, and administrators** to interact through a secure **multi-role** login system. Students can browse, filter, and register for events, while faculty members can create and manage events. Administrators have full control over users, events, and system operations.

The project is developed using **Vibe Coding** a modern web technology including frontend development, and database for backend processing and data management. Additional features such as email notifications, data visualization using charts, and a Multilanguage AI chatbot are integrated to enhance user experience.

INTRODUCTION

In today's digital era, educational institutions are rapidly adopting technology to improve communication, management, and collaboration. Colleges organize various activities such as technical events, cultural programs, workshops, seminars, conferences, and sports competitions, but most of these activities are managed manually or through separate systems. This creates issues like lack of coordination, limited participation, and poor information flow among students and faculty members.

To overcome these challenges, the project "One Stop Solution" is developed as a centralized web-based platform designed specifically for all colleges in Amritsar. The platform provides a unified system where colleges can register, manage their students and staff, and organize events efficiently. It ensures that all activities are available on a single platform, making it easier for users to access information, participate in events, and stay updated with notifications.

➔ Importance of the Website:

The importance of such a platform lies in its ability to simplify and modernize the traditional event management system used in colleges. A centralized website acts as a bridge between multiple institutions, allowing them to collaborate while maintaining their own data privacy. It eliminates the need for manual notices, paperwork, and disconnected communication channels.

This system is especially useful for students, as they can explore events not only from their own college but also from other colleges in Amritsar. It increases participation, exposure, and opportunities for learning and interaction. Faculty members benefit by easily managing event details, tracking participants, and verifying registrations. Administrators gain full control over their college data, ensuring transparency, security, and proper management of all users and activities.

➔ Unique ID Generation and Security System:

One of the most important features of this platform is the automatic generation of unique IDs for different user roles, which ensures security and prevents unauthorized access. When a user creates an account, the system automatically assigns a role-based unique ID that cannot be manually modified by the user.

For example, students are assigned IDs like “STU10001”, while staff members receive IDs like “STF10001”, and administrators are assigned IDs such as “ADM1001”. These ID formats are intentionally designed to be different in structure and pattern so that users cannot access portals that do not belong to their role. This prevents students or staff from attempting to log in to the admin portal using fake credentials.

Additionally, each college has only one authorized admin account, and its ID is uniquely generated when the college registers on the platform. Since the ID format for admin is different from student and staff IDs, it becomes impossible for users to create fake admin accounts. This system strengthens authentication and ensures that only valid and authorized users can access their respective dashboards.

This role-based unique ID system, combined with password authentication and OTP verification, provides a high level of security, transparency, and trust across the platform.

➔ Technologies for this Project:

The development of the “One Stop Solution” platform is based on modern web technologies to ensure high performance, scalability, and a smooth user experience. The frontend of the application is developed using React.js, which provides a dynamic and component-based architecture for building interactive user interfaces. TypeScript is used along with React to improve code quality, maintainability, and error handling, making the application more robust and reliable.

For styling and design, Tailwind CSS is used to create a modern, responsive, and visually appealing interface. It allows the use of utility-first classes to design clean layouts, attractive dashboards, and smooth animations. The project uses Vite as a development tool, which provides fast build times and efficient performance during development.

On the backend side, technologies such as Node.js and Express.js are used to handle server-side logic, API creation, and data processing. The database is managed using MySQL, which stores all important information such as user details, events, registrations, and payments securely.

➔ Overview of System Functionality:

The platform is designed with three main user roles: Student, Staff, and Admin, each having a separate dashboard and specific functionalities. Students can register, manage their profiles, upload ID cards, explore events, and participate in activities. Staff members can create and manage events, verify payments, and monitor student participation. Administrators have full control over their college data, including managing users, approving participants, and maintaining system integrity.

One of the key features of this system is multi-college support with strict data privacy, where each college operates independently. Students from one college can apply for events in another college, but participation is only confirmed after approval by the organizing college's admin.

Literature Review

The literature review focuses on studying the existing systems and technologies related to event management platforms and educational collaboration systems. Many institutions still rely on traditional methods for managing events such as noticeboards, paper-based registrations, and manual record keeping. These methods require a significant amount of time and effort from staff members and often lead to problems such as data loss, miscommunication, and difficulty in managing large numbers of participants.

Several online event management platforms have been developed to simplify the process of organizing and managing events. These systems allow users to create events, manage registrations, and communicate with participants through digital platforms. However, most of these systems are designed for commercial events and are not specifically tailored to the requirements of educational institutions. They do not provide specialized features such as college-based administration, student verification through ID cards, or role-based access for students, faculty members, and administrators.

Educational institutions also use learning management systems to manage academic activities and communication between students and teachers. While these systems are useful for managing courses and assignments, they do not provide effective solutions for managing inter-college events, conferences, workshops, or cultural activities. As a result, there is a need for a dedicated platform that can manage both academic collaborations and event participation between different colleges.

Modern web development technologies have made it possible to build advanced and scalable web applications that can support multiple users and institutions at the same time. Technologies such as React for frontend development, TypeScript for structured programming, and Tailwind CSS for responsive design provide an efficient way to create interactive user interfaces. On the backend side, frameworks like Node.js and database systems like MySQL allow developers to securely manage user information, event data, and transaction records.

Another important aspect identified in previous studies is the need for proper security and authentication mechanisms in online platforms. Many systems implement role-based access control where different users have different permissions based on their roles. For example, students can register for events, staff members can manage events and verify payments, and administrators can monitor and manage the entire system. Secure login methods, unique ID generation, and document verification are also essential to prevent fake registrations and unauthorized access.

Recent advancements in digital platforms also highlight the importance of user experience and intelligent assistance. A well-designed user interface improves usability and ensures that students and staff can easily navigate the system. In addition, the integration of AI-based chatbots can help users by answering common questions, guiding them through the registration process, and providing quick support when needed.

Based on the analysis of existing systems and technologies, it can be concluded that there is a strong need for a centralized platform that can connect multiple colleges while maintaining data privacy and transparency. The proposed “One Stop Solution” platform aims to address these limitations by providing a secure, modern, and user-friendly system where students, staff, and administrators can manage events, workshops, conferences, and other academic activities efficiently.

PROBLEM STATEMENT

1. Lack of Centralized Platform

Before the development of this system, there was no single platform where all colleges could manage and share their events. Each college used its own methods such as notice boards, WhatsApp groups, or separate websites. Because of this, students were not able to get complete information about events happening in other colleges, which reduced their participation and opportunities.

2. Limited Inter-College Participation

Students were mostly restricted to participating in events within their own college. There was no proper system to allow students from different colleges to register and participate in events easily. This lack of connection between colleges limited students' exposure, knowledge sharing, and collaboration with others.

3. Manual Data Management by Staff

Faculty members had to manage all event-related data manually, such as maintaining participant lists, registration details, and attendance records. This process was time-consuming and difficult, especially when the number of participants was large. Managing these records during events became complex and stressful for staff members.

4. Risk of Data Loss

Since most of the data was stored in physical form like papers or registers, there was always a risk of losing important information. Sometimes registration sheets or participant records were misplaced, which created confusion and affected the smooth management of events.

5. Difficulty in Payment Handling

For events that required registration fees, payments were handled manually. Faculty had to verify payments without any proper system, which increased the chances of errors and confusion. There was no clear way to track whether a student's payment was approved or not.

6. Lack of Proper Verification System

There was no strong system to verify whether a student was genuine or not. Without proper identity verification, fake registrations could occur, which affected the credibility of the events and created problems for organizers.

7. No Role-Based Access System

There was no clear separation of roles between students, staff, and administrators. Everyone depended on manual communication, and there was no proper system to define who can manage what. This lack of structure made the overall process less efficient.

8. Poor Communication and Notifications

Students and staff did not receive proper updates about events, approvals, or changes. Important information was often missed due to lack of a proper notification system, which affected participation and coordination.

9. Time-Consuming Process

All the above problems made the entire system slow and inefficient. Both students and staff had to spend a lot of time managing registrations, verifying data, and communicating manually, which reduced productivity.

SYSTEM ANALYSIS

System Analysis is an important phase in software development where the existing system is carefully studied to understand its working, limitations, and user requirements. It helps in identifying the problems faced by users and provides a clear direction for developing a better and more efficient system.

In this project, system analysis focuses on understanding how traditional college event management systems work and what challenges are faced by students, staff, and administrators. Based on this understanding, a new system is designed that improves performance, security, and usability.

➔ Existing System Analysis

In the existing system, most colleges manage events using manual methods such as notice boards, paper registrations, and informal communication channels like WhatsApp groups. These methods are not organized and often create confusion among students.

Faculty members are responsible for maintaining records of participants, attendance, and event details manually or using basic tools like Excel sheets. This process becomes difficult when the number of participants increases.

Additionally, there is no connection between different colleges. Each college works independently, and students do not get proper information about events happening in other institutions.

➔ Problems in Existing System

The existing system has several major problems. One of the biggest issues is manual data handling, which takes a lot of time and effort. Faculty members have to manage large amounts of data, which increases the chances of mistakes.

Another problem is the risk of data loss. Since records are stored on paper or local files, they can be easily lost or damaged. This creates difficulties during event management and participant verification.

There is also no proper system for inter-college participation, which limits student opportunities. Students cannot explore events outside their college, which reduces their exposure and learning experience.

Payment handling is another issue, as it is done manually without proper tracking. This leads to confusion about whether a payment is completed or not.

➔ Proposed System Analysis

The proposed system, “One Stop Solution,” is designed to solve all these problems by providing a centralized digital platform. It connects multiple colleges and allows them to manage their activities efficiently.

In this system, all processes such as event creation, registration, payment verification, and communication are handled online. This reduces manual work and improves accuracy.

The system also supports inter-college participation, where students can register for events in other colleges. However, participation is controlled through admin approval to maintain discipline and proper management.

➔ Requirements Analysis

The system is developed based on the needs of different users. Students need an easy-to-use platform where they can explore events, register, and track their participation status.

Staff members require tools to create events, manage participants, and verify payments without handling manual records. This helps them save time and reduce workload.

Administrators need full control over their college system. They should be able to manage students, staff, and events, and ensure that all data is secure and properly maintained.

➔ Key Features of the Proposed System

The system includes several advanced features to improve efficiency and user experience. It provides a centralized platform where all colleges can operate independently while still being connected.

One of the key features is the unique ID generation system, where each user is assigned a specific ID such as STU for students, STF for staff, and ADM for admin. This ensures security and prevents unauthorized access.

Other features include online event registration, payment proof upload, student ID verification, real-time notifications, and an AI chatbot for user support. The system also has a modern and responsive design that works on all devices.

SYSTEM DESIGN

➔ Identifying User Requirements:

The first step in system analysis is identifying the needs of different users of the system. In this platform, there are mainly three types of users:

1. Students:

Students need a simple and easy interface to view events, register for participation, upload ID cards, and track their participation status. They also require notifications about event updates and approvals.

2. Faculty:

Staff members need access to create and manage events, verify student registrations, approve or reject payments, and send notifications to students. They also need to monitor participant data efficiently.

3. Administrators:

Administrators manage the overall system of their college. They can control students and staff data, verify users, approve inter-college participation, and maintain security. They also ensure that no fake users exist in the system.

Understanding these requirements helps in designing a system that satisfies all users.

➤ **Functional and Non-Functional Requirements:**

A. Functional Requirements

➤ These are the core features of the system:

- Students, staff, and admin can create accounts and log in using unique IDs (STU, STF, ADM).
- Staff can create events like technical, cultural, sports, workshops, and conferences.
- Students can register for events and upload required details and ID proof.
- Students can upload payment proof, and staff can approve or reject it.
- Users receive updates about events, approvals, and announcements.
- Different colleges can operate independently with data privacy.
- Users can ask queries and get instant help.

B. Non-Functional Requirements

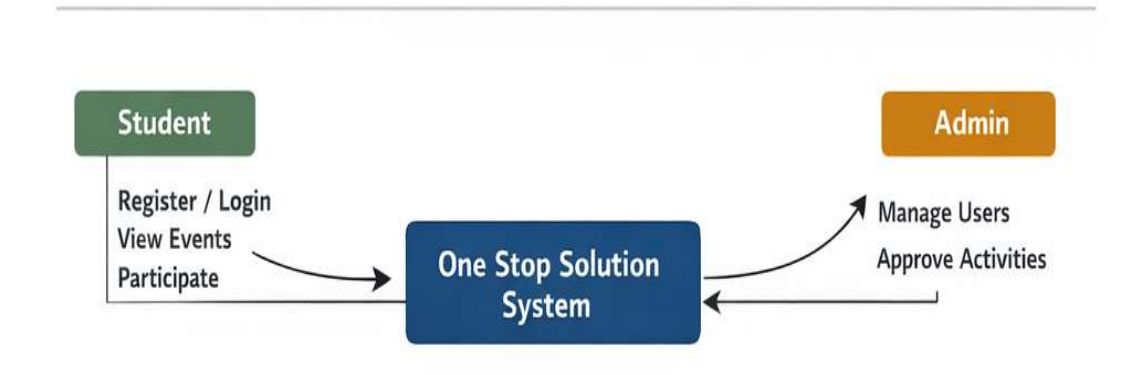
➤ These define system quality:

- The website should load fast and handle multiple users efficiently.
- User data must be protected using authentication, OTP verification, and role-based access.
- The interface should be simple, responsive, and easy to navigate.
- The system should support more colleges and users in the future.

➔ **Data Flow Diagram (DFD):**

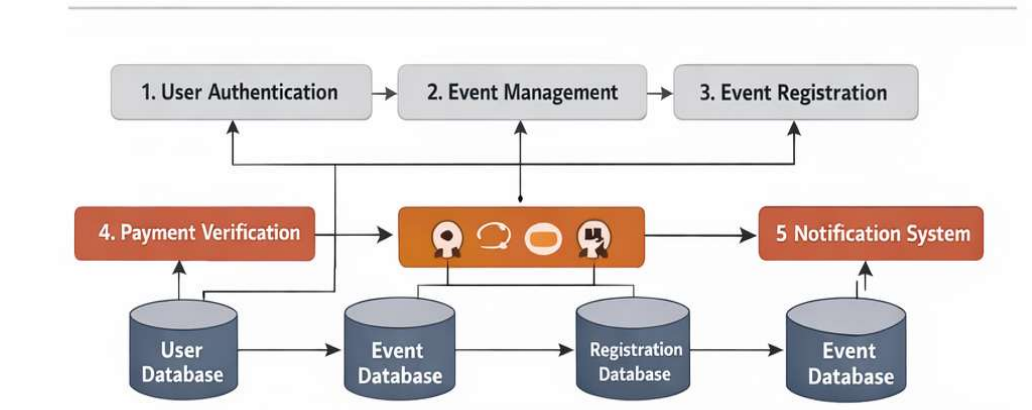
➤ **DFD Level 0:**

DFD Level 0: Context Diagram



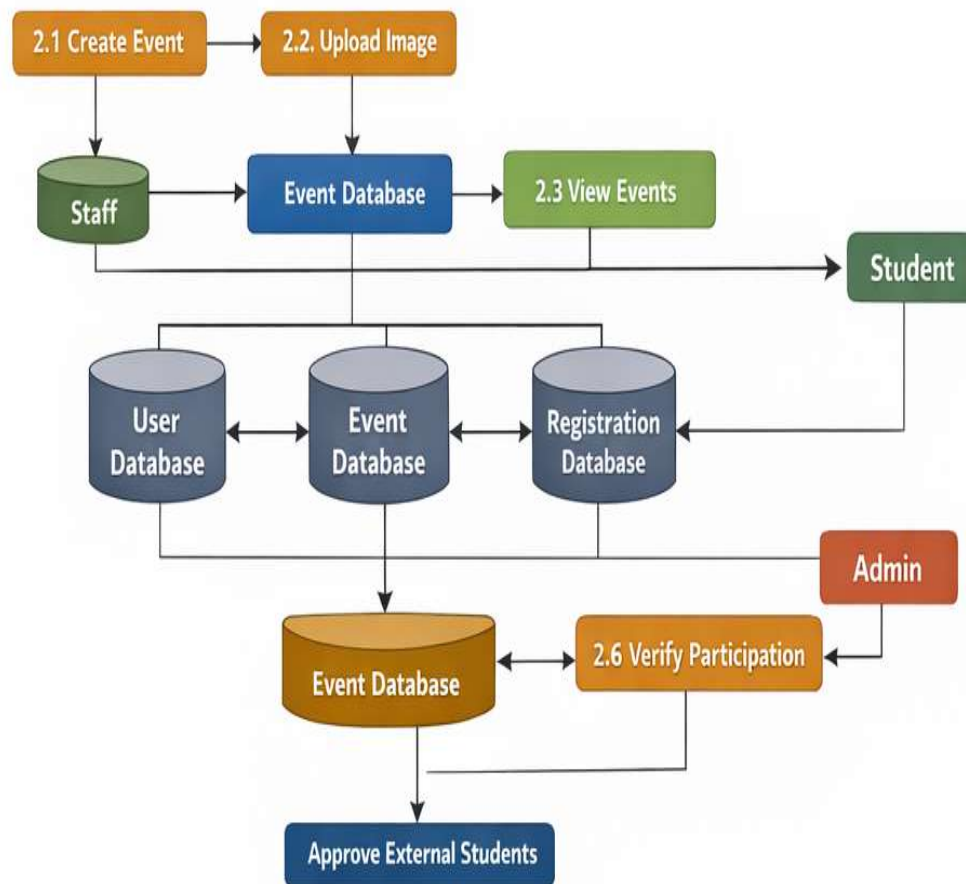
➤ **DFD Level 1:**

DFD Level 1: System Overview



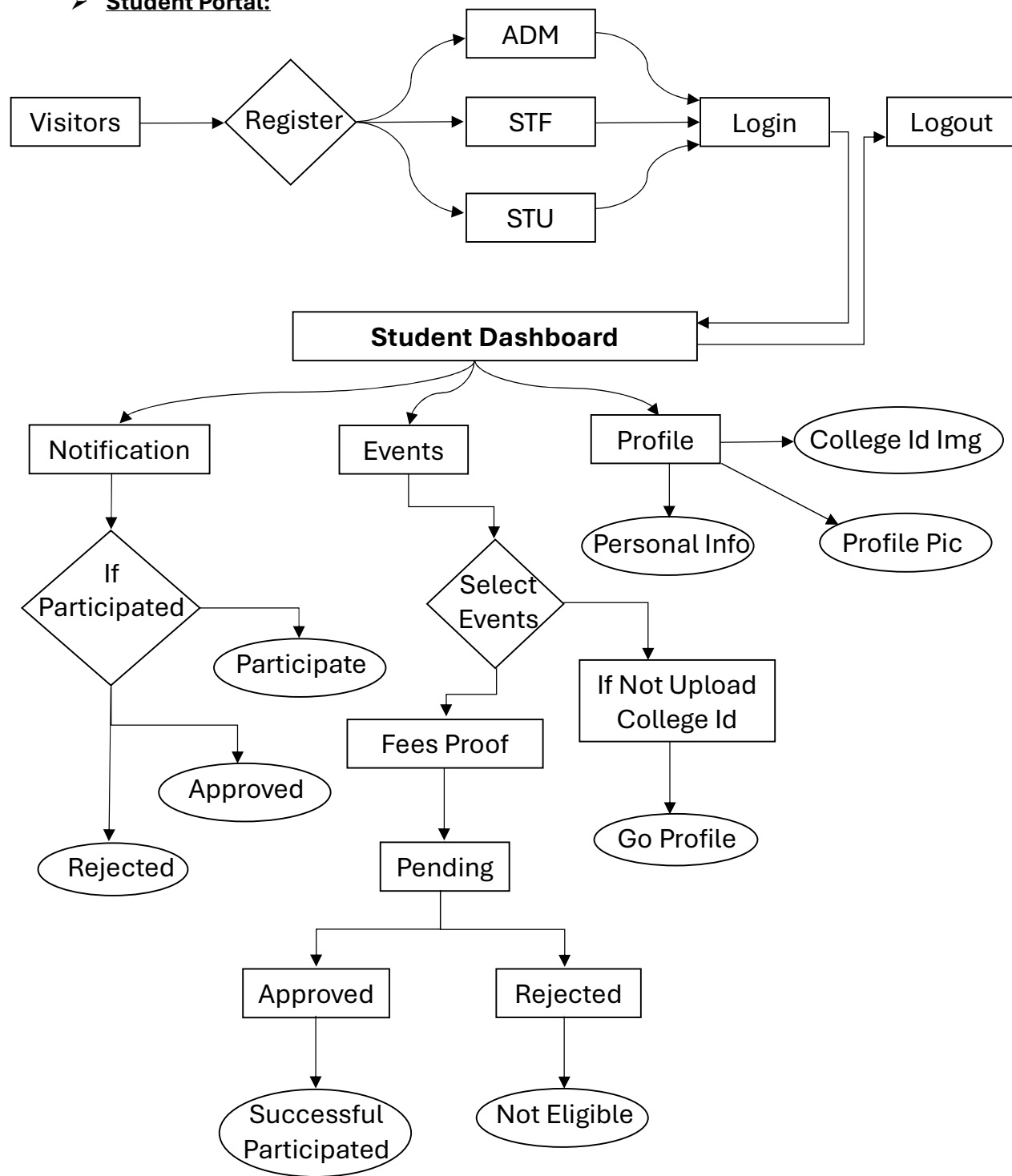
➤ DFD Level 2:

DFD Level 2: Event & Registration Process

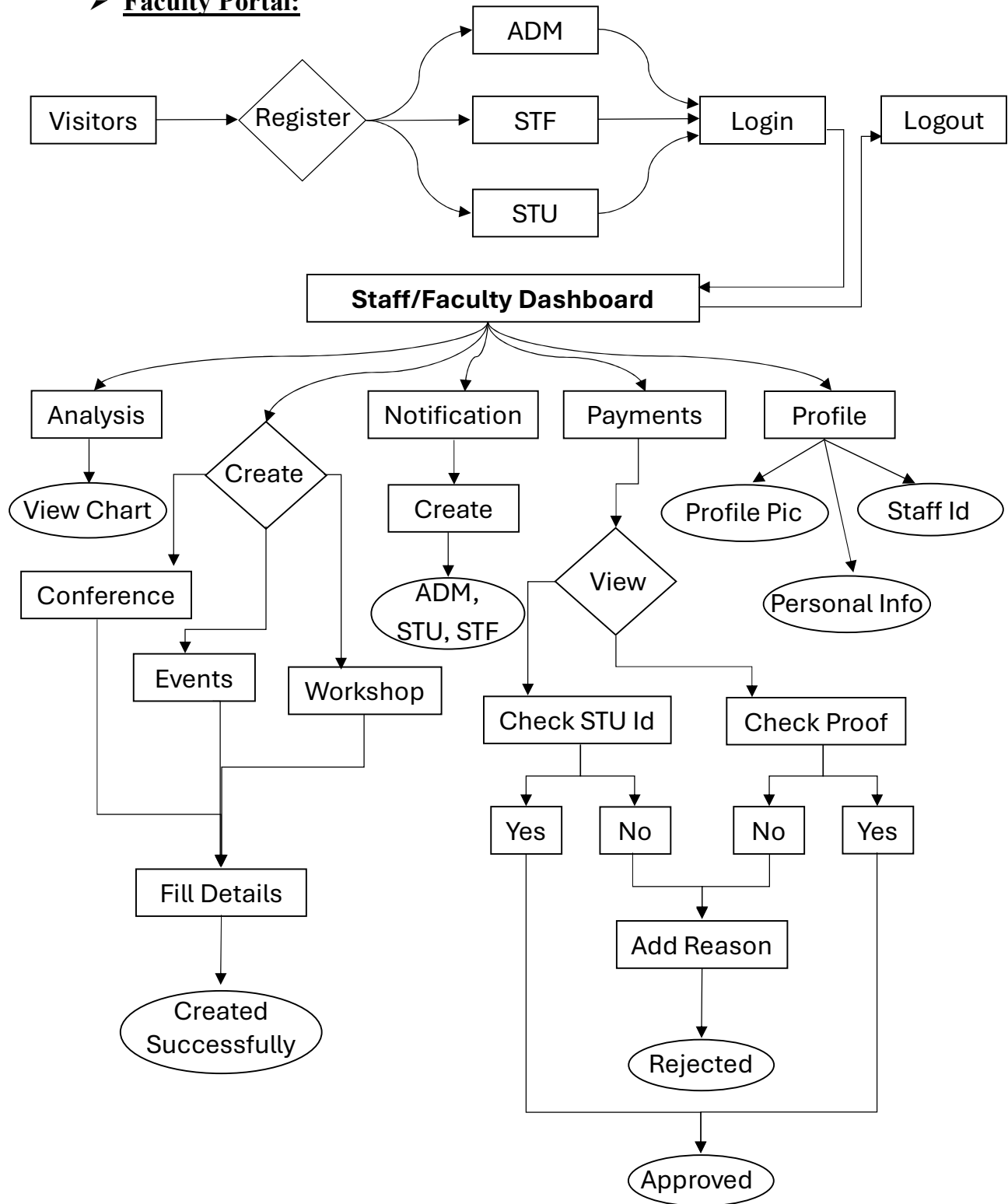


ER. Diagram

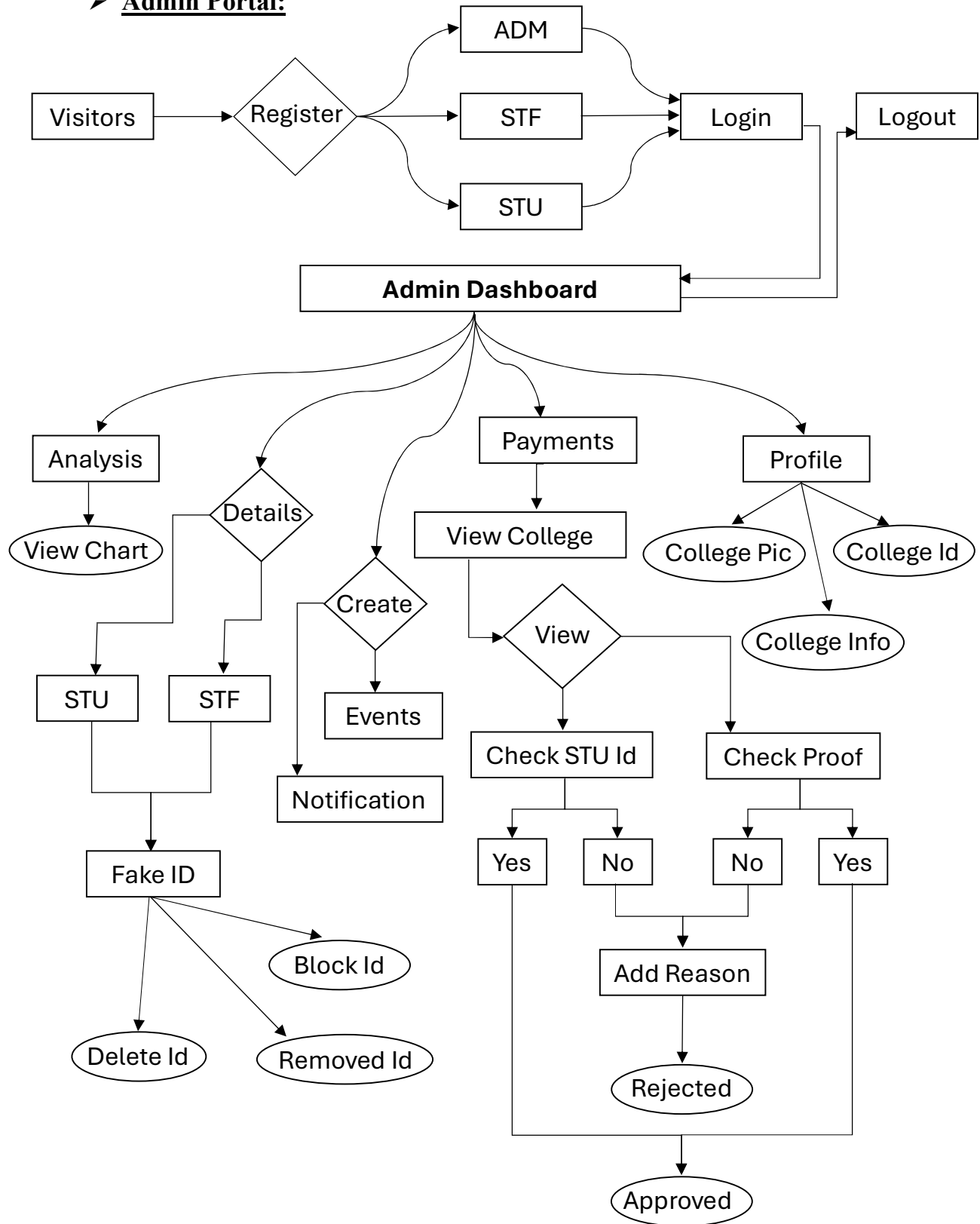
➤ Student Portal:



➤ **Faculty Portal:**



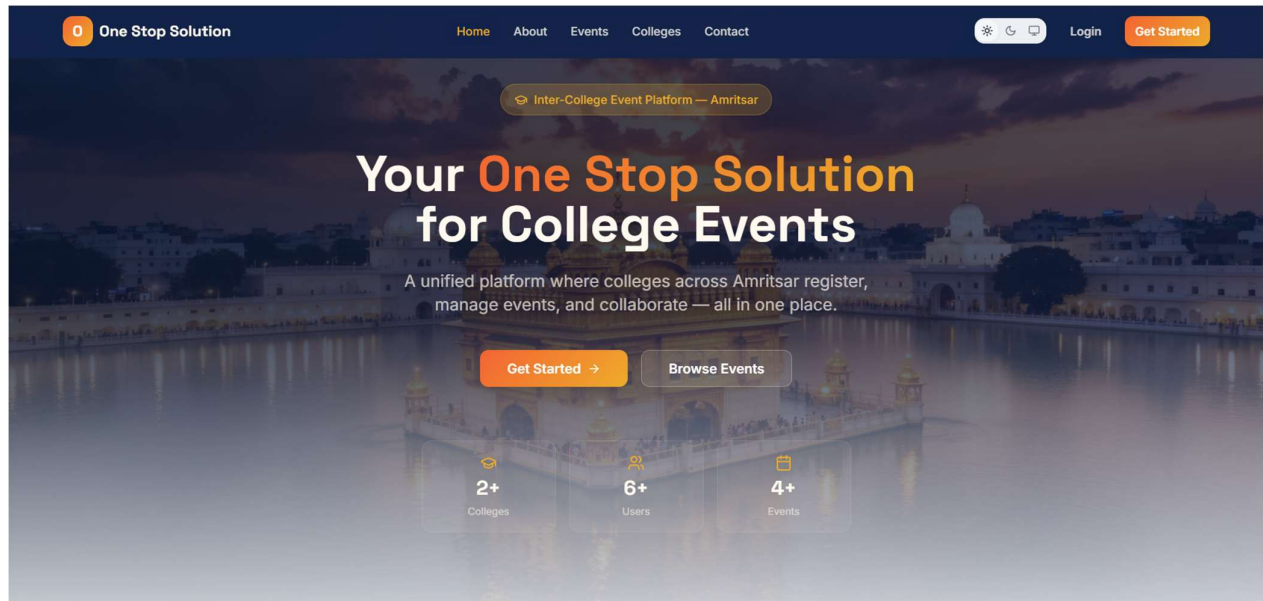
➤ **Admin Portal:**



→ UI / UX Designs:

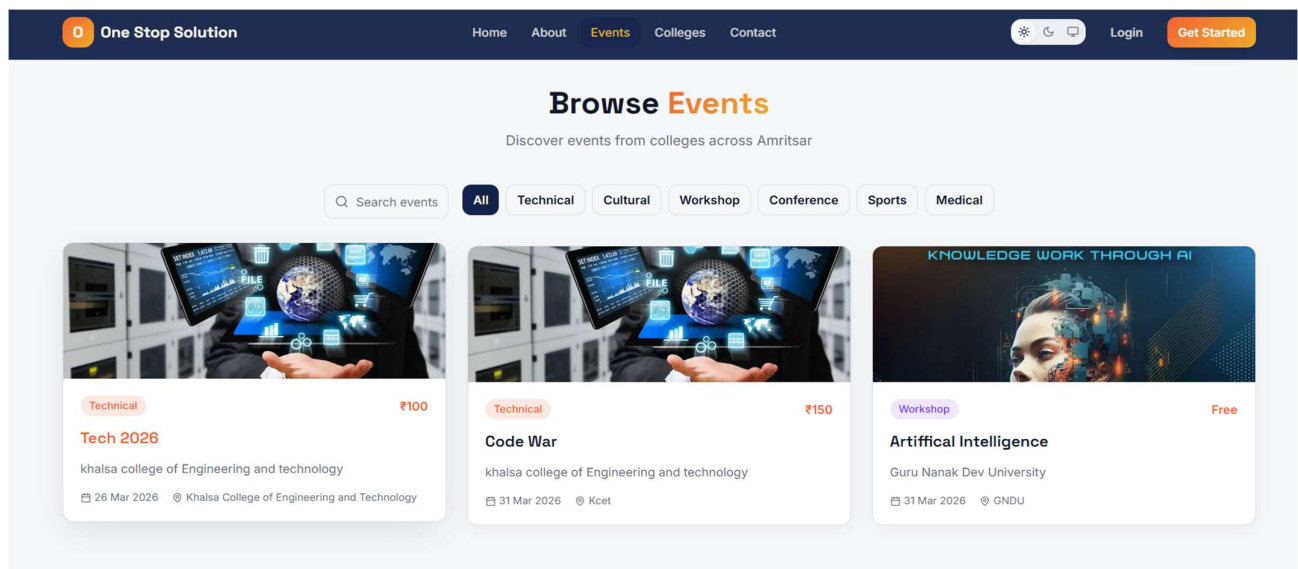
➤ Below are the main sections of the website:

- **Home Section:**



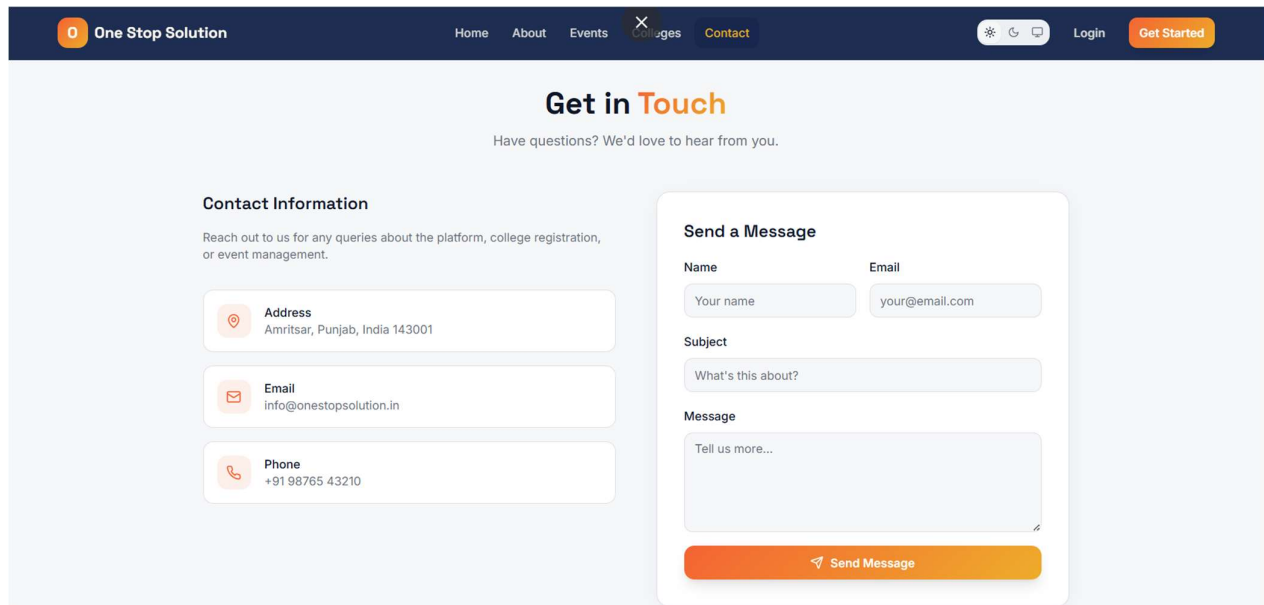
Displays welcome message, navigation bar, and featured events.

- **Events Section:**



Shows all events with details like category, date, and participation.

➤ Contact Section:



The screenshot shows the 'Contact' page of the 'One Stop Solution' website. The header includes the logo, navigation links (Home, About, Events, Colleges, Contact), and utility links (Login, Get Started). The main heading is 'Get in Touch' with the subtext 'Have questions? We'd love to hear from you.' The page is divided into two main sections: 'Contact Information' and 'Send a Message'.

Contact Information

Reach out to us for any queries about the platform, college registration, or event management.

Address
Amritsar, Punjab, India 143001

Email
info@onestopsolution.in

Phone
+91 98765 43210

Send a Message

Name
Your name

Email
your@email.com

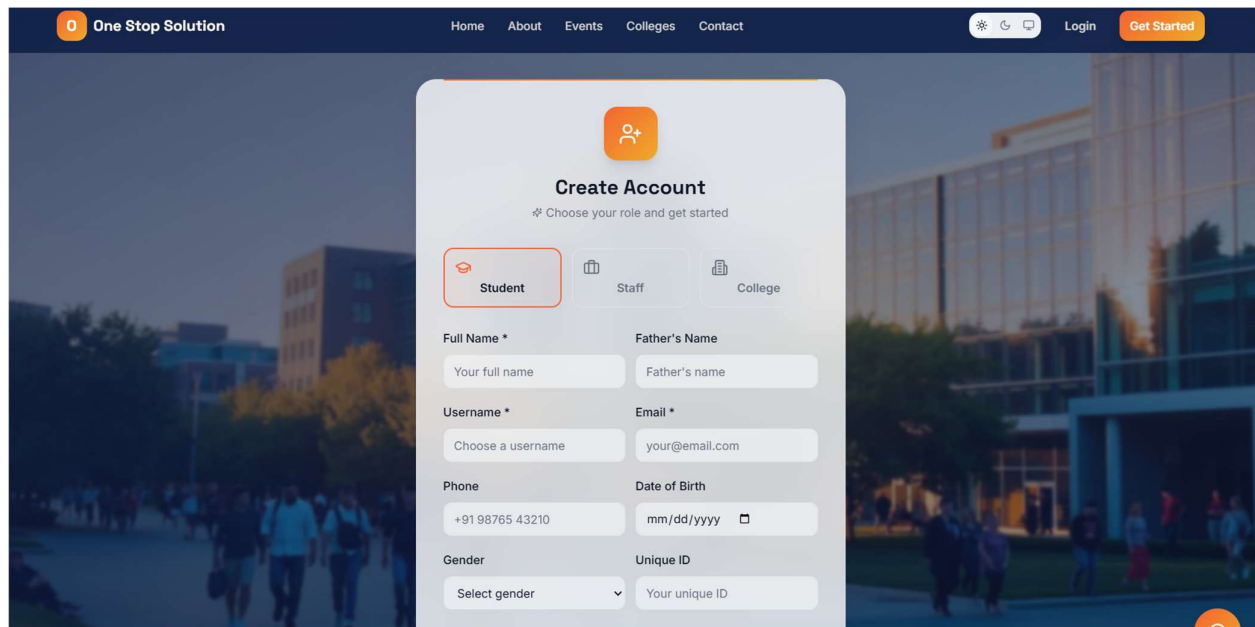
Subject
What's this about?

Message
Tell us more...

Send Message

Allows users to send queries to the system.

• Registration Page:



The screenshot shows the 'Create Account' page of the 'One Stop Solution' website. The header is identical to the contact page. The main heading is 'Create Account' with the subtext 'Choose your role and get started'. There are three role selection buttons: 'Student' (selected), 'Staff', and 'College'. Below the role selection, there are several input fields for user registration.

Create Account
Choose your role and get started

Student **Staff** **College**

Full Name *
Your full name

Father's Name
Father's name

Username *
Choose a username

Email *
your@email.com

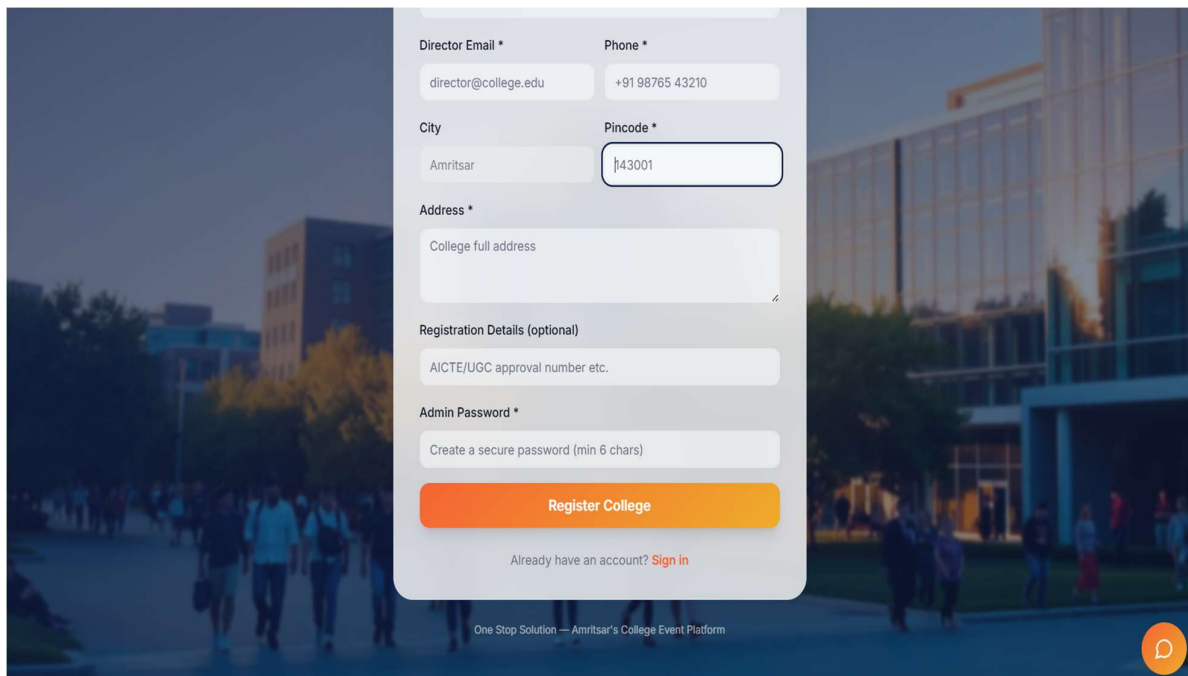
Phone
+91 98765 43210

Date of Birth
mm/dd/yyyy

Gender
Select gender

Unique ID
Your unique ID

Select College *



Registration form for Amritsar's College Event Platform. The form is overlaid on a background image of a modern college building with students walking. The form fields include:

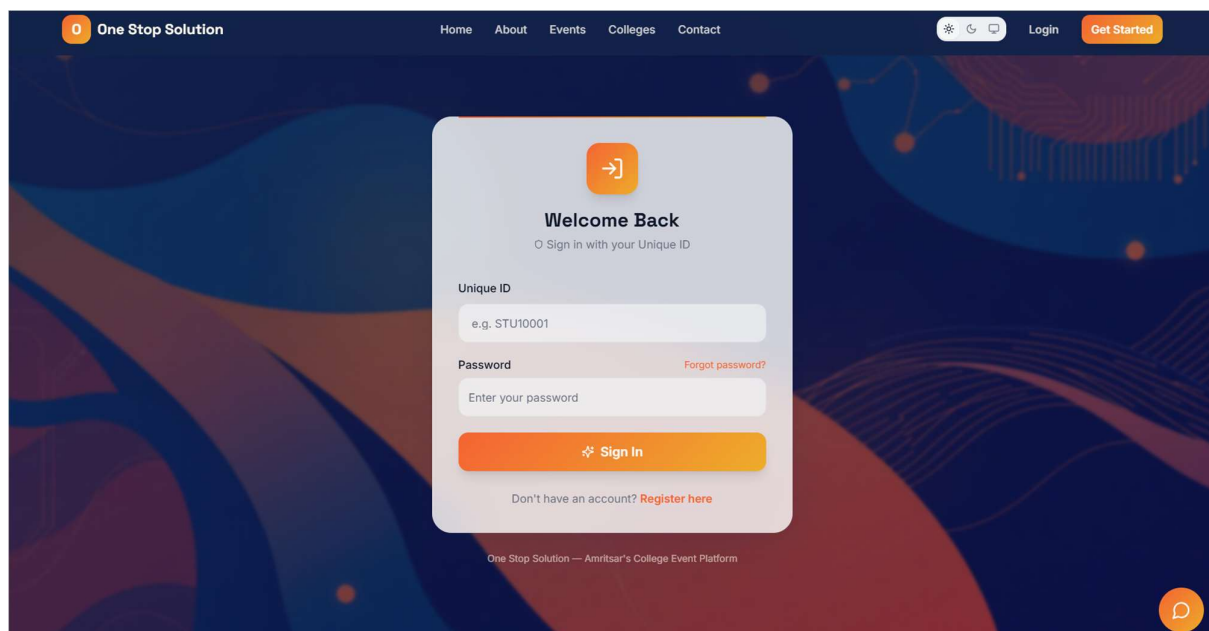
- Director Email *: director@college.edu
- Phone *: +91 98765 43210
- City: Amritsar
- Pincode *: 143001
- Address *: College full address
- Registration Details (optional): AICTE/UGC approval number etc.
- Admin Password *: Create a secure password (min 6 chars)

Buttons: Register College (orange), Already have an account? [Sign in](#) (text link).

Footer: One Stop Solution — Amritsar's College Event Platform

Users can Register as Student, Staff, Admin Separated registrations for all.

- **Signup page:**



Welcome Back login page for Amritsar's College Event Platform. The page features a dark blue background with abstract orange and white patterns. The login form is centered and includes:

- Header: One Stop Solution (orange button), Home, About, Events, Colleges, Contact, Login, Get Started (orange button).
- Form Fields: Unique ID (e.g. STU10001), Password (Enter your password).
- Buttons: Sign In (orange), Register here (text link).

Footer: One Stop Solution — Amritsar's College Event Platform

User can sign up with their Unique Id user got after the registration.

Software & Hardware Requirements:

➤ **Hardware Requirements:**

S. No.	Component	Specification
1	Processor	Intel i3 or above
2	RAM	Minimum 4 GB (8 GB recommended)
3	Storage	500 GB HDD / 256 GB SSD
4	Device	Tablet/ Desktop / Smartphone
5	Internet Connection	Mobile Data / Wi-Fi

➤ **Software Requirements:**

S. No.	Software/Tool	Description
1	React	Frontend development for User Interface
2	TypeScript	Strong typing scripts for better code quality
3	Tailwind CSS	Styling and responsive design of the overall Pages
4	Node.js	Backend runtime environment
5	Operating System	Windows, Mac, Linux
6	SQL/Supabase	Database management system
7	Vite	Fast development and build tool
8	Web Browser	Chrome, Edge, Firefox for running the web application

IMPLEMENTATION

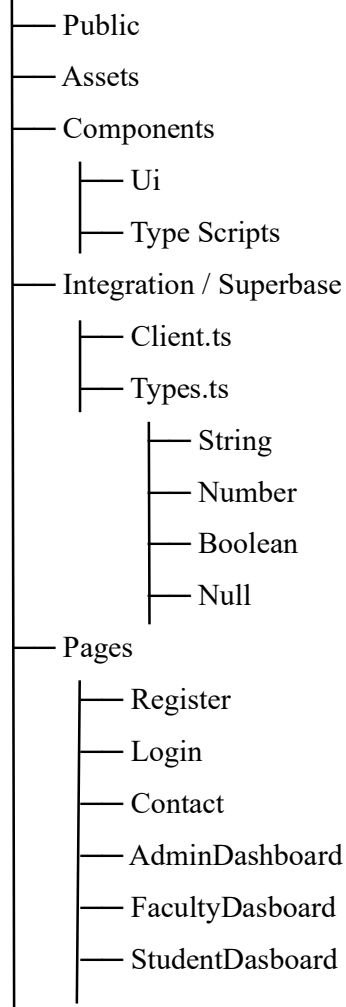
➔ SYSTEM ARCHITECTURE OVERVIEW

The **One Stop Solution** platform is implemented using a **full-stack architecture using vite coding**, where the frontend, backend, and database are separated to ensure scalability and maintainability.

➔ PROJECT STRUCTURE

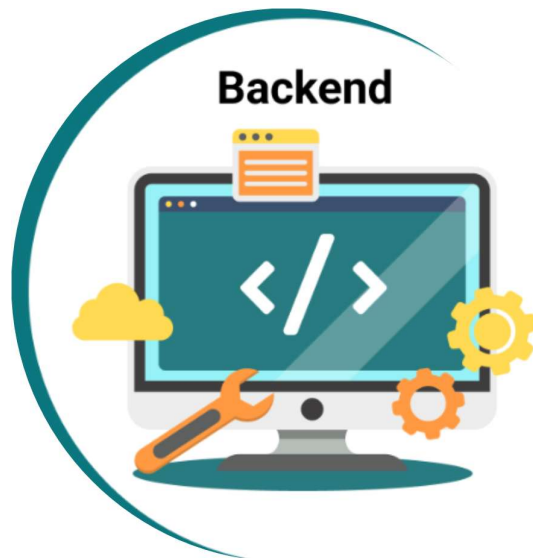
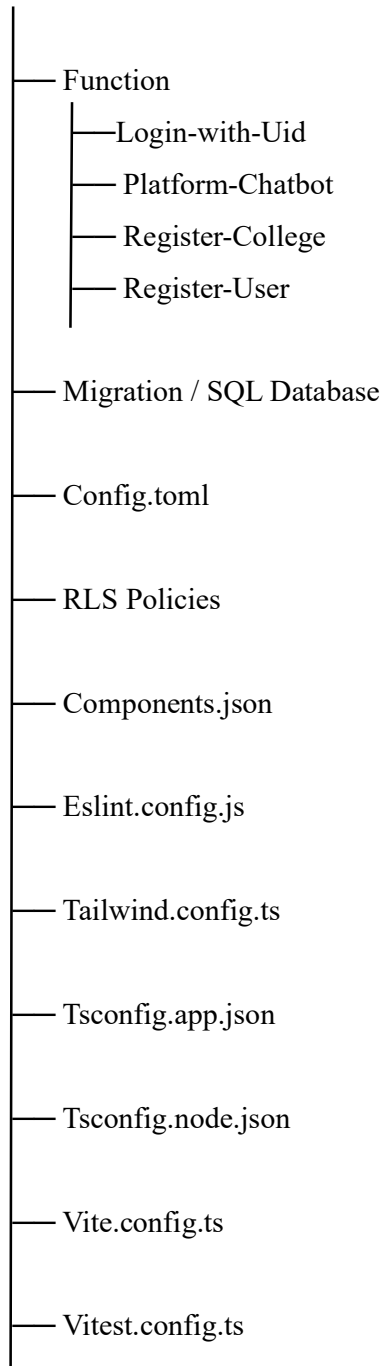
➤ Frontend Structure

/Frontend



➤ Backend Structure

/Backend



➔ FRONTEND IMPLEMENTATION

➤ UI:

➤ Avatar.tsx:

```
import * as React from "react";
import * as AvatarPrimitive from "@radix-ui/react-avatar";

import { cn } from "@lib/utils";

const Avatar = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Root>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Root
    ref={ref}
    className={cn("relative flex h-10 w-10 shrink-0 overflow-hidden rounded-
full", className)}
    {...props}
  />
));
Avatar.displayName = AvatarPrimitive.Root.displayName;

const AvatarImage = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Image>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Image>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Image ref={ref} className={cn("aspect-square h-full w-
full", className)} {...props} />
));
AvatarImage.displayName = AvatarPrimitive.Image.displayName;

const AvatarFallback = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Fallback>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Fallback>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Fallback
    ref={ref}
    className={cn("flex h-full w-full items-center justify-center rounded-full
bg-muted", className)}
    {...props}
  />
));
AvatarFallback.displayName = AvatarPrimitive.Fallback.displayName;
export { Avatar, AvatarImage, AvatarFallback };
```

➤ Checkbox.tsx:

```
import * as React from "react";
import * as CheckboxPrimitive from "@radix-ui/react-checkbox";
import { Check } from "lucide-react";

import { cn } from "@lib/utils";

const Checkbox = React.forwardRef<
  React.ElementRef<typeof CheckboxPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof CheckboxPrimitive.Root>
>(({ className, ...props }, ref) => (
  <CheckboxPrimitive.Root
    ref={ref}
    className={cn(
      "peer h-4 w-4 shrink-0 rounded-sm border border-primary
      ring-offset-background data-[state=checked]:bg-primary data-
      [state=checked]:text-primary-foreground focus-visible:outline-
      none focus-visible:ring-2 focus-visible:ring-ring focus-
      visible:ring-offset-2 disabled:cursor-not-allowed
      disabled:opacity-50",
      className,
    )}
    {...props}
  >
    <CheckboxPrimitive.Indicator className={cn("flex items-center
    justify-center text-current")}>
      <Check className="h-4 w-4" />
    </CheckboxPrimitive.Indicator>
  </CheckboxPrimitive.Root>
));
Checkbox.displayName = CheckboxPrimitive.Root.displayName;

export { Checkbox };
```

➤ Hover-card.tsx:

```
import * as React from "react";
import * as HoverCardPrimitive from "@radix-ui/react-hover-card";

import { cn } from "@lib/utils";

const HoverCard = HoverCardPrimitive.Root;

const HoverCardTrigger = HoverCardPrimitive.Trigger;

const HoverCardContent = React.forwardRef<
  React.ElementRef<typeof HoverCardPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof
HoverCardPrimitive.Content>
>(({ className, align = "center", sideOffset = 4, ...props },
ref) => (
  <HoverCardPrimitive.Content
    ref={ref}
    align={align}
    sideOffset={sideOffset}
    className={cn(
      "z-50 w-64 rounded-md border bg-popover p-4 text-popover-
foreground shadow-md outline-none data-[state=open]:animate-in
data-[state=closed]:animate-out data-[state=closed]:fade-out-0
data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-
[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2
data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-
in-from-left-2 data-[side=top]:slide-in-from-bottom-2",
      className,
    )}
    {...props}
  />
));
HoverCardContent.displayName =
HoverCardPrimitive.Content.displayName;

export { HoverCard, HoverCardTrigger, HoverCardContent };
```


➤ Scroll-area.tsx

```
import * as React from "react";
import * as ScrollAreaPrimitive from "@radix-ui/react-scroll-area";

import { cn } from "@lib/utils";

const ScrollArea = React.forwardRef<
  React.ElementRef<typeof ScrollAreaPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof ScrollAreaPrimitive.Root>
>(({ className, children, ...props }, ref) => (
  <ScrollAreaPrimitive.Root ref={ref} className={cn("relative overflow-hidden", className)} {...props}>
    <ScrollAreaPrimitive.Viewport className="h-full w-full rounded-[inherit]">{children}</ScrollAreaPrimitive.Viewport>
    <ScrollBar />
    <ScrollAreaPrimitive.Corner />
  </ScrollAreaPrimitive.Root>
));
ScrollArea.displayName = ScrollAreaPrimitive.Root.displayName;

const ScrollBar = React.forwardRef<
  React.ElementRef<typeof
ScrollAreaPrimitive.ScrollAreaScrollbar>,
  React.ComponentPropsWithoutRef<typeof
ScrollAreaPrimitive.ScrollAreaScrollbar>
>(({ className, orientation = "vertical", ...props }, ref) => (
  <ScrollAreaPrimitive.ScrollAreaScrollbar
    ref={ref}
    orientation={orientation}
    className={cn(
      "flex touch-none select-none transition-colors",
      orientation === "vertical" && "h-full w-2.5 border-l border-l-transparent p-[1px]",
      orientation === "horizontal" && "h-2.5 flex-col border-t border-t-transparent p-[1px]",
      className,
    )}
    {...props}
  >
    <ScrollAreaPrimitive.ScrollAreaThumb className="relative flex-1 rounded-full bg-border" />
  </ScrollAreaPrimitive.ScrollAreaScrollbar>
));
```

➤ Admin Dashboard:

```
const AdminDashboard = () => (  
  <DashboardLayout role="admin">  
    <Routes>  
      <Route index element={<Overview />} />  
      <Route path="students" element={<StudentsPage />} />  
      <Route path="staff" element={<StaffPage />} />  
      <Route path="events" element={<EventsPage />} />  
      <Route path="participants" element={<ParticipantsPage />} />  
      <Route path="payments" element={<AdminPayments />} />  
      <Route path="notifications" element={<AdminNotifications />} />  
      <Route path="analytics" element={<Analytics />} />  
      <Route path="profile" element={<CollegeProfile />} />  
    </Routes>  
  </DashboardLayout>  
)  
);
```

➤ Staff Dashboard:

```
const StaffDashboard = () => (  
  <DashboardLayout role="staff">  
    <Routes>  
      <Route index element={<Overview />} />  
      <Route path="events" element={<MyEvents />} />  
      <Route path="create-event" element={<CreateEvent />} />  
      <Route path="participants" element={<Participants />} />  
      <Route path="payments" element={<Payments />} />  
      <Route path="notifications" element={<StaffNotifications />} />  
      <Route path="profile" element={<StaffProfile />} />  
    </Routes>  
  </DashboardLayout>  
)  
);
```

➤ Student Dashboard:

```
// --- Main Dashboard ---
const StudentDashboard = () => (
  <DashboardLayout role="student">
    <Routes>
      <Route index element={<Overview />} />
      <Route path="events" element={<EventsBrowse />} />
      <Route path="registrations" element={<Registrations />} />
      <Route path="profile" element={<Profile />} />
      <Route path="notifications" element={<Notifications />} />
    </Routes>
  </DashboardLayout>
);

export default StudentDashboard;
```

➔ **BACKEND IMPLEMENTATION**

➤ LoginWith-Uid:

```
{
  const { identifier, password } = await req.json()

  if (!identifier || !password) {
    return new Response(
      JSON.stringify({ error: 'Identifier and password are required' }),
      { status: 400, headers: { ...corsHeaders, 'Content-Type':
'application/json' } }
    )
  }
  const supabaseAdmin = createClient(
    Deno.env.get('SUPABASE_URL')!,
    Deno.env.get('SUPABASE_SERVICE_ROLE_KEY')!
  )
  // Look up user by username or unique_id
  const { data: profile, error: lookupError } = await supabaseAdmin
    .from('profiles')
    .select('email')
    .or(`username.eq.${identifier},unique_id.eq.${identifier.toUpperCase(
)})`)
    .maybeSingle()
```

```

    if (lookupError || !profile) {
      return new Response(
        JSON.stringify({ error: 'Invalid username or ID' }),
        { status: 404, headers: { ...corsHeaders, 'Content-Type':
'application/json' } }
      )
    }

    // Sign in with found email
    const supabaseAuth = createClient(
      Deno.env.get('SUPABASE_URL')!,
      Deno.env.get('SUPABASE_ANON_KEY')!
    )

    const { data, error: signInError } = await
supabaseAuth.auth.signInWithPassword({
      email: profile.email,
      password,
    })

    if (signInError) {
      return new Response(
        JSON.stringify({ error: 'Invalid password' }),
        { status: 401, headers: { ...corsHeaders, 'Content-Type':
'application/json' } }
      )
    }

    return new Response(
      JSON.stringify({ session: data.session, user: data.user }),
      { headers: { ...corsHeaders, 'Content-Type': 'application/json' } }
    )
  }
}

```

➤ **PlatformChatbot:**

```

try {
  const { messages, language } = await req.json();
  const LOVABLE_API_KEY = Deno.env.get("LOVABLE_API_KEY");
  if (!LOVABLE_API_KEY) throw new Error("AI service not configured");
  const systemMessages = [{ role: "system", content: SYSTEM_PROMPT }];
  if (language && language !== "en") {

```

```

    systemMessages.push({
      role: "system",
      content: `The user has selected "${language}" as their preferred
language. Always respond in this language.`,
    });
  }
  const response = await
fetch("https://ai.gateway.lovable.dev/v1/chat/completions", {
  method: "POST",
  headers: {
    Authorization: `Bearer ${LOVABLE_API_KEY}`,
    "Content-Type": "application/json",
  },
  body: JSON.stringify({
    model: "google/gemini-3-flash-preview",
    messages: [...systemMessages, ...messages],
    stream: true,
  }),
});
if (!response.ok) {
  if (response.status === 429) {

```

```

    return new Response(JSON.stringify({ error: "Too many requests.
Please wait a moment." }), {
    status: 429, headers: { ...corsHeaders, "Content-Type":
"application/json" },
    });
  }
  if (response.status === 402) {
    return new Response(JSON.stringify({ error: "Service credits
exhausted." }), {
    status: 402, headers: { ...corsHeaders, "Content-Type":
"application/json" },
    });
  }
  const t = await response.text();
  console.error("AI error:", response.status, t);
  return new Response(JSON.stringify({ error: "AI service error" }), {
    status: 500, headers: { ...corsHeaders, "Content-Type":
"application/json" },
    });
  }
  return new Response(response.body, {
    headers: { ...corsHeaders, "Content-Type": "text/event-stream" },
  });
}

```

➔ MIGRATION / SQL DATABASE IMPLEMENTATION

➤ User Table:

```
-- User roles table
CREATE TABLE public.user_roles (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES auth.users(id) ON DELETE CASCADE NOT NULL,
  role public.app_role NOT NULL,
  UNIQUE (user_id, role)
);
```

➤ Events Registration:

```
-- Event registrations
CREATE TABLE public.event_registrations (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  event_id UUID REFERENCES public.events(id) ON DELETE CASCADE NOT NULL,
  student_id UUID REFERENCES auth.users(id) ON DELETE CASCADE NOT NULL,
  registration_status public.registration_status NOT NULL DEFAULT 'pending',
  payment_status public.payment_status NOT NULL DEFAULT 'pending',
  payment_proof_url TEXT,
  rejection_reason TEXT,
  requires_admin_approval BOOLEAN NOT NULL DEFAULT false,
  admin_approved BOOLEAN,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  UNIQUE(event_id, student_id)
);
```

➤ **Profile Table:**

-- Profiles table

```
CREATE TABLE public.profiles (  
  id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,  
  unique_id TEXT UNIQUE NOT NULL,  
  username TEXT UNIQUE NOT NULL,  
  full_name TEXT NOT NULL,  
  father_name TEXT,  
  email TEXT NOT NULL,  
  phone TEXT,  
  dob DATE,  
  gender public.gender_type,  
  address TEXT,  
  college_id UUID REFERENCES public.colleges(id),  
  department TEXT,  
  roll_number TEXT,  
  avatar_url TEXT,  
  college_id_card_url TEXT,  
  is_active BOOLEAN NOT NULL DEFAULT true,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

➤ **Row-Level Security (RLS) Policies:**

- Row-Level Security (RLS) is a database security feature that allows restricting access to specific rows in a table based on the user's identity or role. It ensures that users can only view or modify the data that they are authorized to access.
- RLS is commonly used in systems where multiple users share the same database but should not have access to each other's data, such as multi-college platforms, banking systems, and enterprise applications.

- Colleges

2 Policies

<u>Policy name</u>	<u>Command</u>	<u>Applies to</u>	<u>Rule expression</u>
<u>Admins can update own college</u>	<u>UPDATE</u>	<u>authenticated</u>	<u>Using((id = get user college(auth.uid())) AND has_role(auth.uid(), 'admin'::app_role))</u>
<u>Anyone can view active colleges</u>	<u>SELECT</u>	<u>public</u>	<u>Using(is_active = true)</u>

- User Roles

3 Policies

<u>Policy name</u>	<u>Command</u>	<u>Applies to</u>	<u>Rule expression</u>
<u>Admins can view college user roles</u>	<u>SELECT</u>	<u>authenticated</u>	<u>Using(has_role(auth.uid(), 'admin'::app_role) AND (user_id IN (SELECT profiles.id FROM profiles WHERE (profiles.college_id = get_user_college(auth.uid())))))</u>
<u>Staff can view college user roles</u>	<u>SELECT</u>	<u>authenticated</u>	<u>Using(has_role(auth.uid(), 'staff'::app_role) AND (user_id IN (SELECT profiles.id FROM profiles WHERE (profiles.college_id = get_user_college(auth.uid())))))</u>
<u>Users can view own roles</u>	<u>SELECT</u>	<u>authenticated</u>	<u>Using(user_id = auth.uid())</u>

TESTING

Testing is the process of checking whether the system is working properly or not. It helps to find errors and fix them before the final use. In this project, testing is done to make sure that all features like login, event creation, registration, and payment system are working correctly.

➔ TEST CASES

➤ Test Case 1: Login System

Test Case ID	TC01
Description	Verify user login
Input	Valid email & password
Expected Output	Redirect to dashboard
Result	Pass

➤ Test Case 2: Payment Upload

Test Case ID	TC02
Description	Upload payment proof

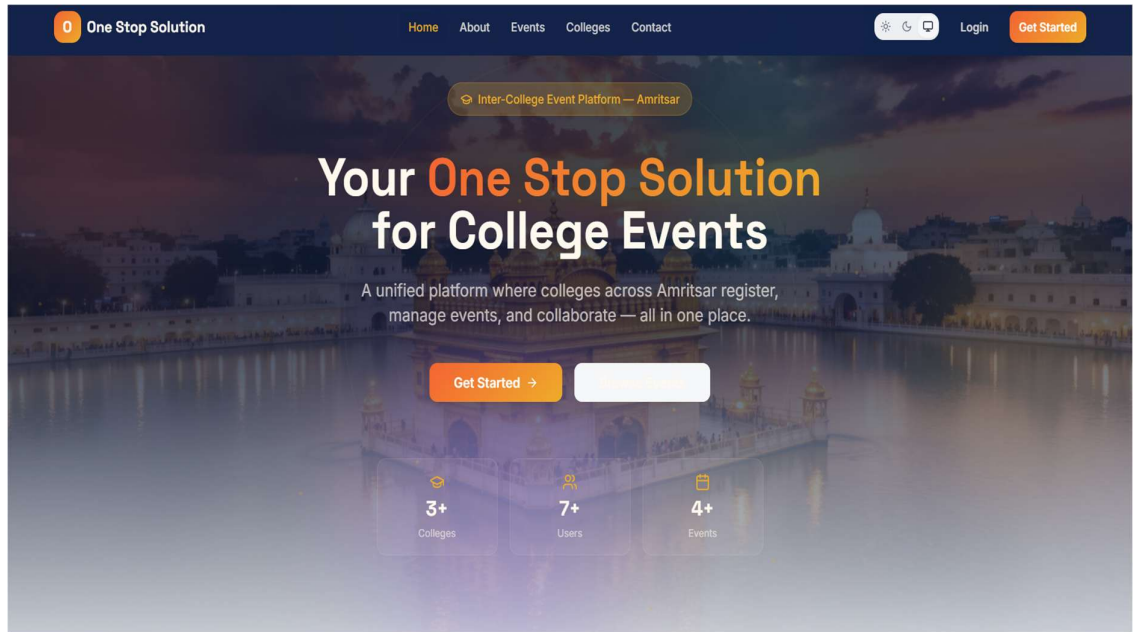
Input	Image file
Expected Output	File uploaded
Result	Pass

➤ **Test Case 3: Admin Control**

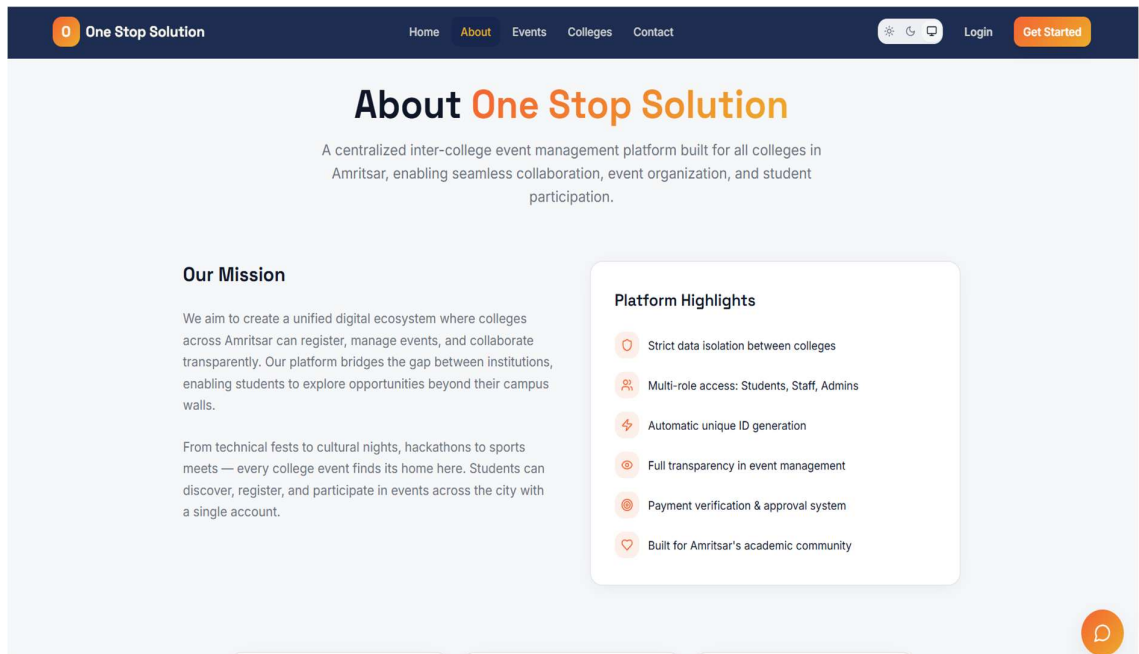
Test Case ID	TC03
Description	Admin manages users
Input	User actions
Expected Output	User updated/deleted
Result	Pass

RESULT / DISCUSSION

➤ Home Section:



➤ About Section:



➤ Events Section:

The screenshot shows the 'Events' section of the 'One Stop Solution' website. The header includes navigation links: Home, About, Events (highlighted), Colleges, and Contact. There are also icons for a sun, a clock, and a chat bubble, along with 'Login' and 'Get Started' buttons. The main heading is 'Browse Events' with the subtitle 'Discover events from colleges across Amritsar'. Below this is a search bar 'Search events' and a category filter bar with buttons: All (selected), Technical, Cultural, Workshop, Conference, Sports, and Medical. Three event cards are displayed:

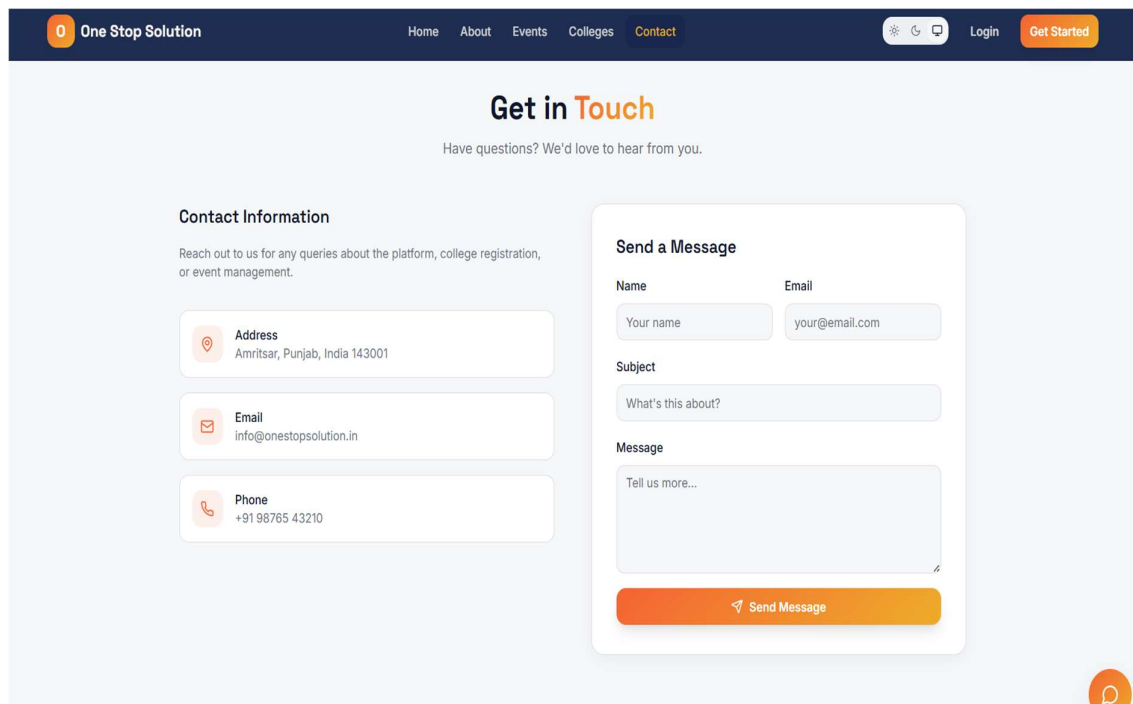
- Tech 2026**: Technical event, ₹100, hosted by Khalsa College of Engineering and Technology on 26 Mar 2026.
- Code War**: Technical event, ₹150, hosted by Khalsa College of Engineering and Technology on 31 Mar 2026.
- Artificial Intelligence**: Workshop, Free, hosted by Guru Nanak Dev University on 31 Mar 2026.

➤ Colleges Section:

The screenshot shows the 'Colleges' section of the 'One Stop Solution' website. The header is identical to the Events section. The main heading is 'Registered Colleges' with the subtitle 'Institutions partnered with One Stop Solution in Amritsar'. Below this is a search bar 'Search colleges...'. Three college cards are displayed:

- DAV College**: Code DC1003, GNDU. Address: DAV COLLEGE Katra Sher Singh I/S Hathi Gate AMRITSAR. Phone: 6383537000. Email: directormail123@gmail.com.
- Guru Nanak Dev University**: Code GNDU1002, GNDU. Address: Grand Trunk (GT) Road, Near Naya Gaon, Amritsar, A... Phone: 2345678912. Email: gndudirector@gmail.com.
- Khalsa College of Engineering and Technology**: Code KCOE1001, ptu. Address: c block, Ranjit avenue, Amritsar, Amritsar - 143001. Phone: 1234567891. Email: xyz123@gmail.com.

➤ Contact Section:



The screenshot shows the 'Contact' page of the 'One Stop Solution' website. The header includes the logo, navigation links (Home, About, Events, Colleges, Contact), and utility links (Login, Get Started). The main heading is 'Get in Touch' with the subtext 'Have questions? We'd love to hear from you.' The page is divided into two main sections: 'Contact Information' and 'Send a Message'.

Contact Information

Reach out to us for any queries about the platform, college registration, or event management.

Address
Amritsar, Punjab, India 143001

Email
info@onestopsolution.in

Phone
+91 98765 43210

Send a Message

Name
Your name

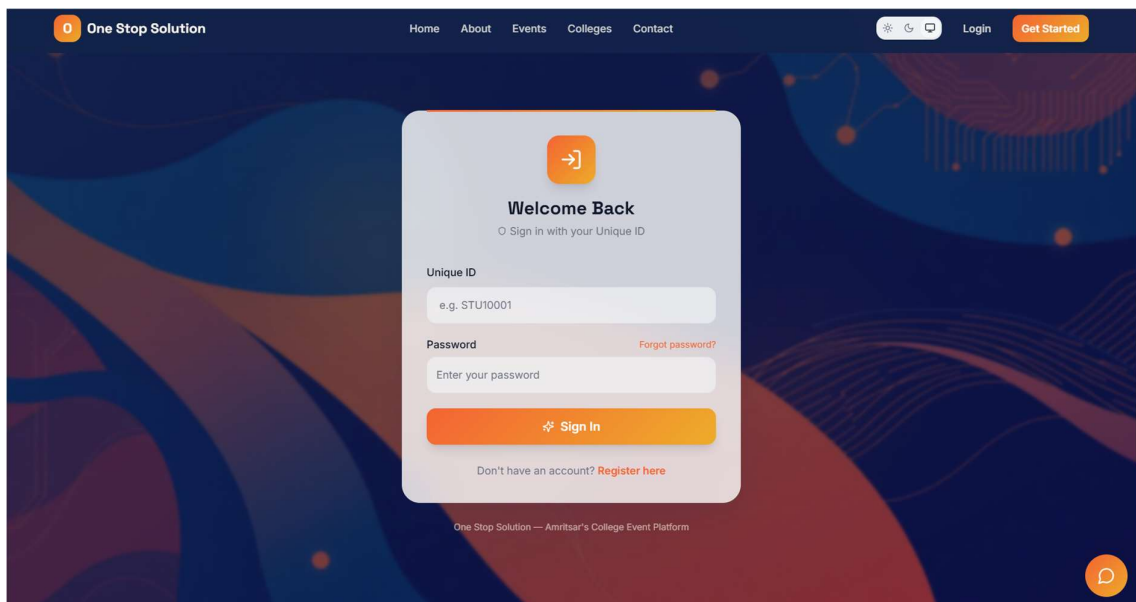
Email
your@email.com

Subject
What's this about?

Message
Tell us more...

Send Message

➤ Login Section:



The screenshot shows the 'Login' page of the 'One Stop Solution' website. The header is identical to the contact page. The main heading is 'Welcome Back' with the subtext 'Sign in with your Unique ID'. The login form is centered and includes fields for 'Unique ID' and 'Password'. A 'Forgot password?' link is next to the password field. A 'Sign In' button is at the bottom of the form. A link to 'Register here' is provided for users who don't have an account. The footer text reads 'One Stop Solution — Amritsar's College Event Platform'.

Welcome Back
Sign in with your Unique ID

Unique ID
e.g. STU10001

Password
Enter your password

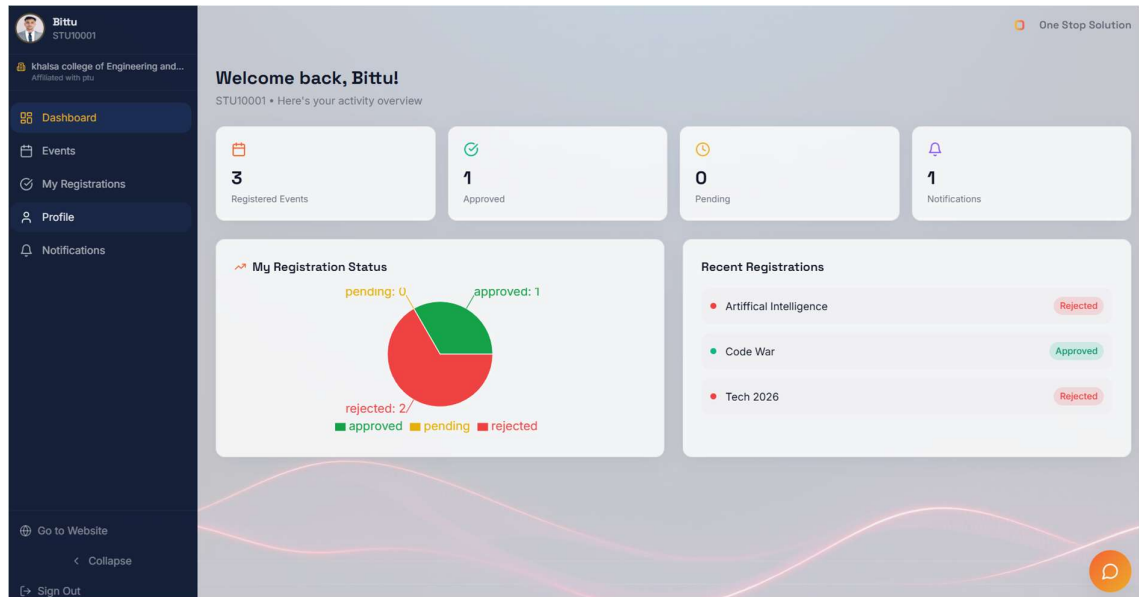
[Forgot password?](#)

Sign In

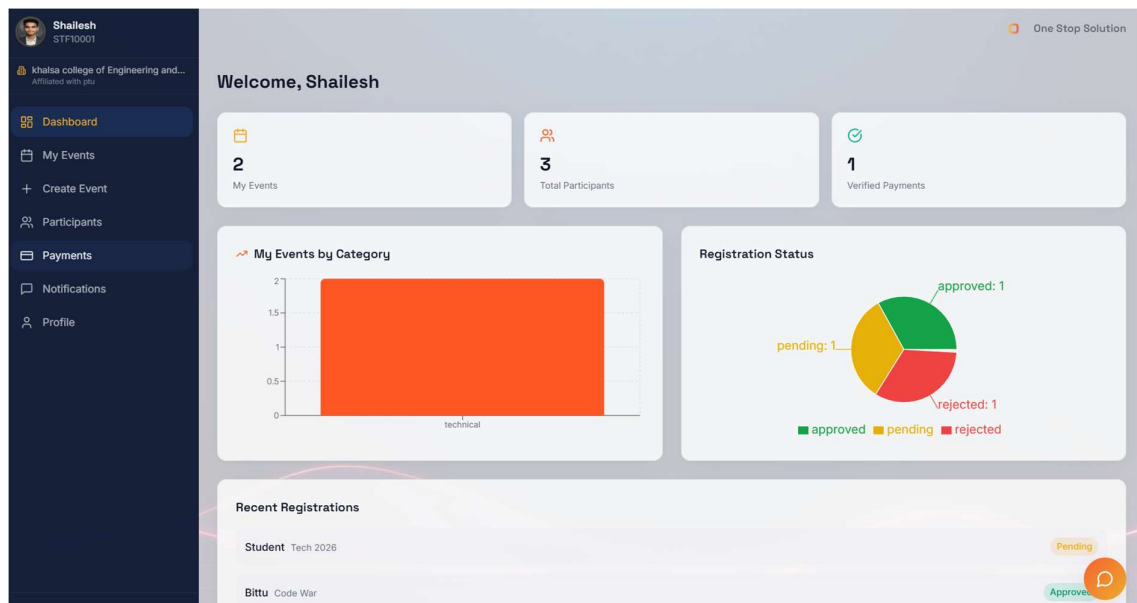
Don't have an account? [Register here](#)

One Stop Solution — Amritsar's College Event Platform

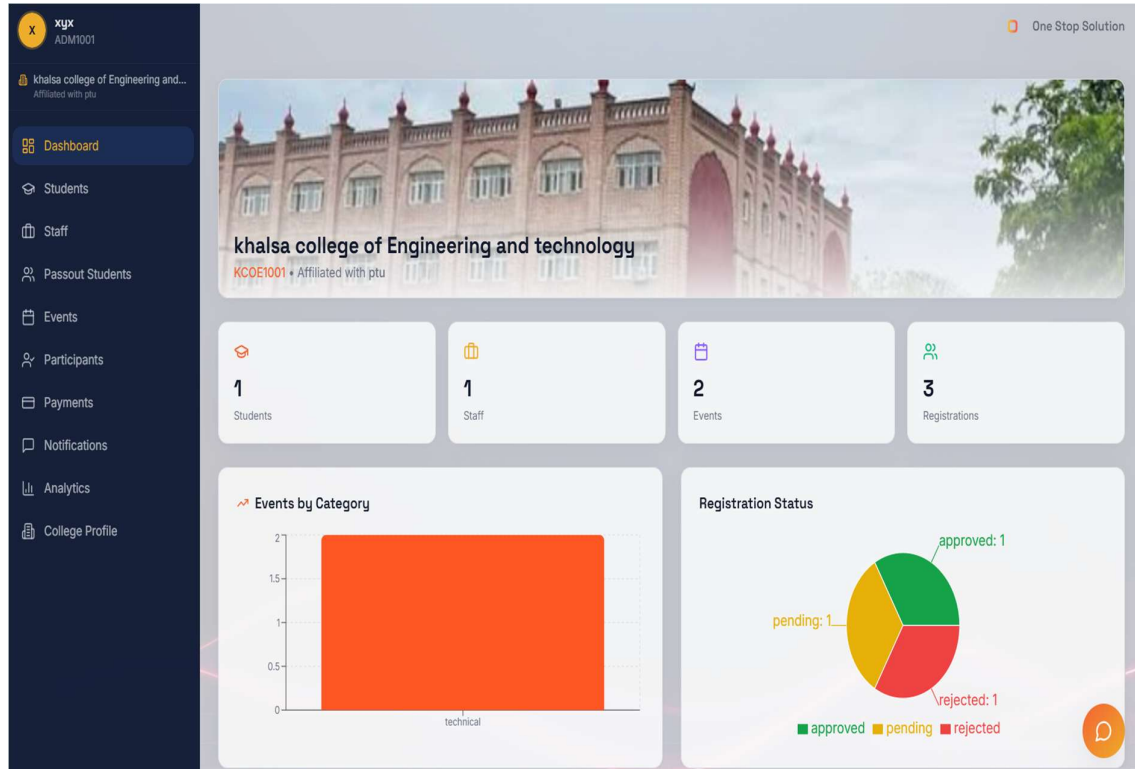
➤ Student Dashboard:



➤ Faculty Dashboard:



➤ **Admin Dashboard:**



Future Scope

1. Functional

1.1 E-Certificate Management

The platform allows efficient management of students, staff, and administrators, where some other college students need the e-certificate.

2. Multi-College

2.1 College Integration

The system supports multiple colleges with around in the India Colleges, allowing each college to register and operate independently.

2.2 Advanced Data Privacy and Transparency

Each college has its own secure database, and no other college can access its internal data. This ensures privacy while maintaining transparency in participation.

3.2 Efficient Event Handling

All event-related tasks such as registration, approval, and notifications are handled digitally, saving time for both students and staff.

4. Student Scope

4.1 Knowledge and Exposure

The platform allows students to interact with participants from other colleges, which helps in gaining knowledge, improving skills, and building confidence.

4.2 Collaboration and Networking

Students can build relationships with peers from different institutions through team participation and events, which enhances their communication and teamwork skills.

5. Faculty Scope

5.1 Participant Tracking

The system provides a structured way to track student participation, making it easier to monitor event success and engagement.

6. Inter-College Collaboration Scope

6.1 Cross-College Interaction

Students from different colleges can participate in shared events, improving collaboration between institutions.

6.2 Controlled Participation

Participation is approved by the organizing college's admin, ensuring proper management and discipline.

CONCLUSION

The **“One Stop Solution”** project successfully provides a modern, centralized platform for managing inter-college events, workshops, and activities in an efficient and organized manner. It overcomes the limitations of traditional manual systems by introducing digital features such as online registration, secure authentication, payment verification, and real-time notifications. The system ensures proper role-based access for students, staff, and administrators, making management easier and more structured.

The platform not only reduces time and manual effort but also improves communication and transparency within colleges. It enables students to participate in events across different institutions, helping them gain knowledge, exposure, and collaboration opportunities. With its modern UI/UX design, responsive layout, and secure backend, the system provides a user-friendly and reliable experience.

Overall, the project achieves its goal of creating an efficient, scalable, and secure solution for inter-college event management. It has strong potential for future enhancements and expansion beyond Amritsar, making it a valuable and practical system in the field of education and technology.

REFERENCES

➔ **The Following of the given the projects references are the Mentioned Below:**

- Pressman, R. S. Software Engineering: A Practitioner's Approach, McGraw-Hill.
- Sommerville, I. Software Engineering, Pearson Education.
- MDN Web Docs – <https://developer.mozilla.org/>□
- W3Schools – <https://www.w3schools.com/>□
- React Official Documentation – <https://react.dev/>□
- Node.js Documentation – <https://nodejs.org/>□
- MySQL Documentation – <https://dev.mysql.com/doc/>□
- Tailwind CSS Documentation – <https://tailwindcss.com/>□
- Chart.js Documentation – <https://www.chartjs.org/>□
- Firebase (OTP Authentication) – <https://firebase.google.com/>□

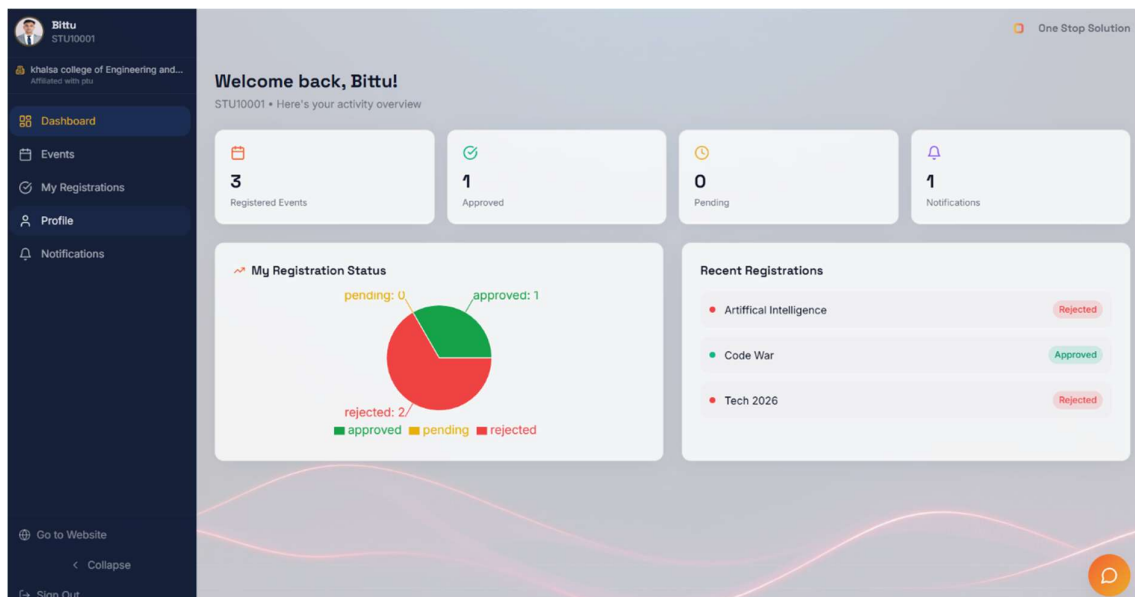
APPENDIX

The appendix section contains additional supporting information related to the “One Stop Solution” project, which helps in better understanding the implementation and working of the system. This section includes important code snippets, screenshots of different modules, database structure, and additional features that are not included in the main report.

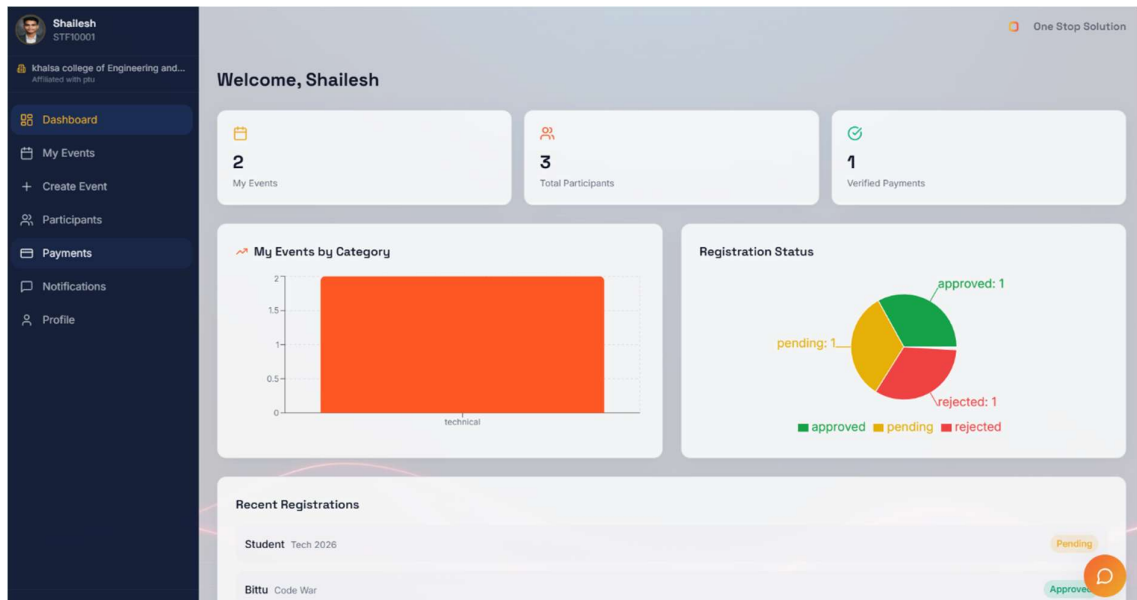
➔ Screenshots of the System

This section includes screenshots of various pages of the developed web application such as the Student Dashboard, Staff Dashboard, Admin Dashboard, Events Page, and Notification Section. These screenshots help in visualizing the user interface and overall design of the system.

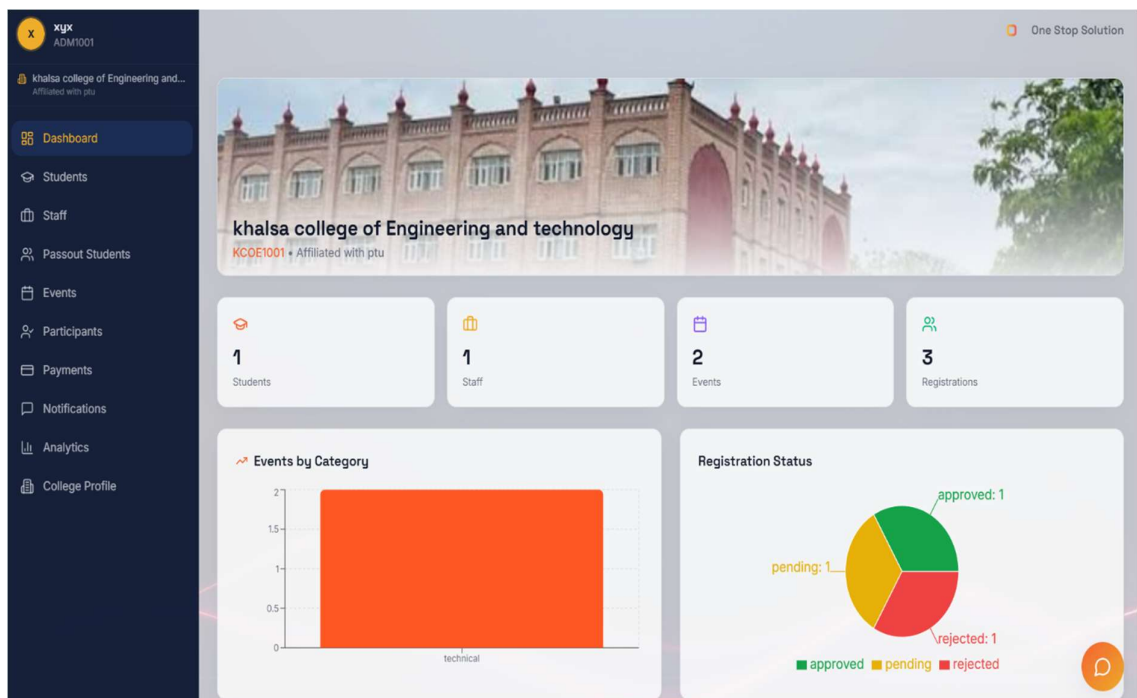
➤ Student Portal:



➤ **Faculty Portal:**

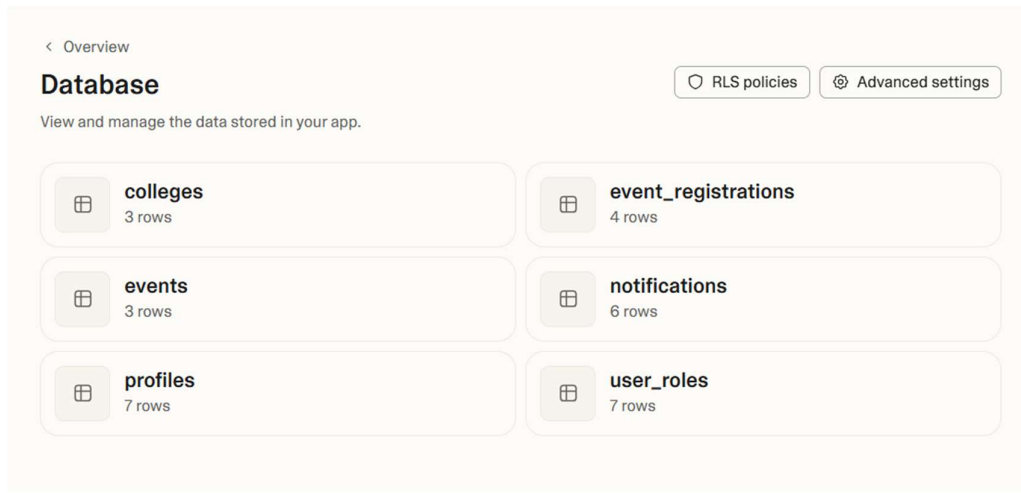


➤ **Admin Portal:**



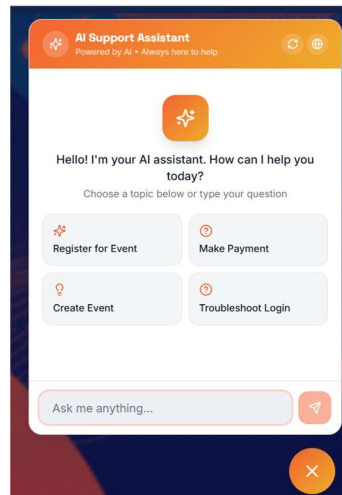
➔ Backend System:

The database used in this project is tables are properly structured to see the overall backend, users increase the traffic see & overall.



➔ Additional Features:

The project includes several additional features such as AI chatbot support.



➔ Project Resources:

This section may include the project source code link (GitHub) resources of this project for further reference.

“<https://github.com/aditya57657/amritsar-connect>”